

Beginner's Guide to ER Diagram Creation (Train Station Management Example)

Overview

This handout explains how to build an **Entity–Relationship (ER) diagram** step by step, following the conventions used in **Database System Concepts** by Korth and Silberschatz. We'll design a small database for a ****Train Station Management System**** — starting from the scenario, identifying entities, attributes, and relationships, and then combining everything into a complete ER diagram.

1. Understanding the Scenario

Scenario: A train station database keeps track of trains, stations, and passengers. Each train has a unique train number, a name, and a type (e.g., express, local). Each station has a unique code, name, and location. A train stops at multiple stations according to a schedule. Passengers can book tickets for a specific train between a source and destination station.

- Entities: `Train`, `Station`, `Passenger`, `Ticket`
- Relationships:
 - `Stops_At` — between `Train` and `Station`
 - `Books` — between `Passenger` and `Ticket`
 - `Includes` — between `Ticket` and `Train`

2. Step 1 – Identify Entities and Attributes

Each entity is shown as a rectangle, and its attributes as ellipses.

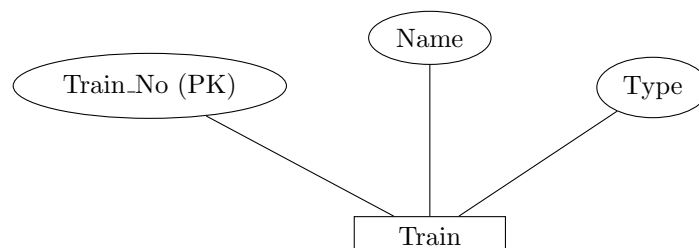


Figure 1: Entity: `Train` with attributes.

Similarly:

- **Station:** Station_Code (PK), Name, Location
- **Passenger:** Passenger_ID (PK), Name, Age
- **Ticket:** Ticket_No (PK), Date, Fare, Source, Destination

3. Step 2 – Define Relationships

(a) Stops_At Relationship

A train stops at many stations, and each station can have many trains stopping there. This is a **many-to-many (M:N)** relationship.



Figure 2: M:N relationship: Train–Stops_At–Station.

This relationship might include attributes such as **Arrival_Time** and **Departure_Time**.

(b) Books Relationship

A passenger can book many tickets, but each ticket is booked by one passenger. This is a **one-to-many (1:N)** relationship.



Figure 3: 1:N relationship: Passenger–Books–Ticket.

(c) Includes Relationship

Each ticket includes one train, and a train can appear in many tickets. This is a **one-to-many (1:N)** relationship from Train to Ticket.

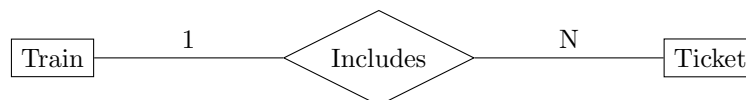


Figure 4: 1:N relationship: Train–Includes–Ticket.

4. Step 3 – Add Relationship Attributes (if any)

Some relationships carry additional information. For example, **Stops_At** can have:

- Arrival_Time
- Departure_Time
- Platform_No

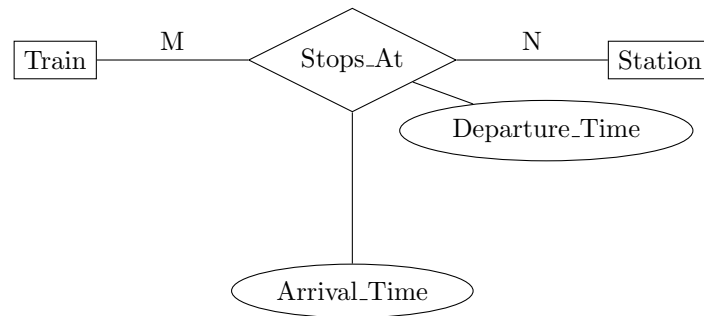


Figure 5: Relationship attributes added to Stops_At.

5. Step 4 – Combine into the Final ER Diagram

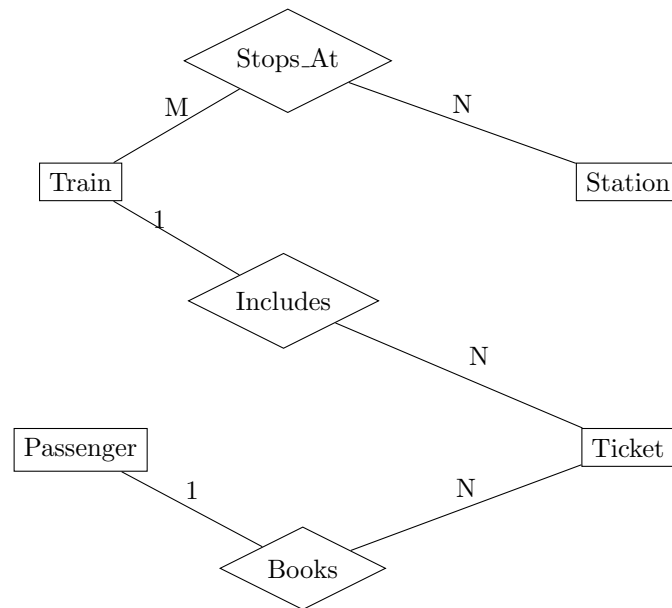


Figure 6: Final ER Diagram: Train Station Management System.

6. Recap and Design Insights

- **Entities** capture main objects — Train, Station, Passenger, Ticket.
- **Relationships** capture how entities interact — Stops_At, Includes, Books.
- Use **primary keys (PK)** to uniquely identify each entity.
- Add relationship attributes (like times, platform number) only when needed.
- For many-to-many relationships, you may later convert them to a separate table (e.g., Stops_At table).

Further Reading

Silberschatz, Korth, and Sudarshan, *Database System Concepts*, 7th Edition — Chapters on ER modeling and database design.